



दक्षिण बिहार केन्द्रीय विश्वविधालय
Central University of South Bihar

Question Paper of End-Term Examination-Dec. 2025

Name of Department: Computer Science

Programme: M.Sc(AI)

Session: 2024-26

Semester: III

Course Title: Artificial Neural Network and Deep Learning Course Code: CAI91DC01804

Course Credit: 4

Duration: 3 Hours

Enrollment Number:

Max Marks: 70

There are three sections (A, B, and C). Answer all questions as per instructions. Points against each question are mentioned in the parenthesis. Use of a calculator is not allowed

Section - A

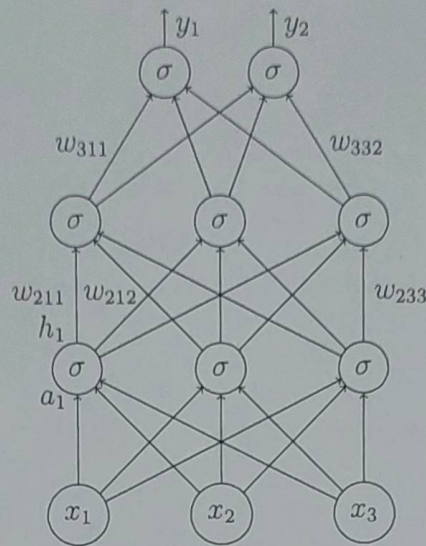
Answer all the questions in not more than 50 words. Each question carries 1 Points.
(1 × 10 = 10 Points)

1. Write the update rule of the gradient descent algorithm.
2. Define McCulloch Pitts neuron function.
3. Depends on the output, which output activation function, and the loss function we use.
4. What is a contour map?
5. Write the update rule of Adgrad.
6. State the link between PCA and autoencoder.
7. Plot the graph between model complexity and error for training error and test error.
8. Define ReLU function.
9. Describe one hot representation of words with an example.
10. Write the full form of CNN, RNN, LSTM, and GRU.

Section - B

Answer ANY FIVE questions in not more than 300 words. Each question carries 6 Points.
(6 × 5 = 30 Points)

1. Mathematically show that the error on the test data gives the more clear picture of the error of a learned function in comparison to the error on the training data.
2. Suppose in a deep neural network (See the figure in the second page) that takes input $x = (x_1, \dots, x_n)$ (layer 0), W_{ijk} represents weight of the link (j, k) at layer i where the node j is present at layer $i - 1$ and the node k is at layer i , $a_i = (\sum_{j=1}^n W_{ij1}h_{j1}, \dots, \sum_{j=1}^n W_{ijn}h_{jn})$ represents the input vector at layer i , $h_i = (\sigma(a_{i1}), \dots, \sigma(a_{in}))$ represents the output vector of n neurons present at layer i for $i \neq 0$, for $i = 0$, $h_0 = (x_1, \dots, x_n)$. At the output layer, we have two neurons and the output we represent as a vector y . Note that $1 \leq i, j, k \leq n$. Show that $\nabla_{W_k} \mathcal{L}(\theta) = \nabla_{W_k} \mathcal{L}(\theta) h_{k-1}^T$ or derive the formula for $\nabla_{a_3} \mathcal{L}(\theta) = \frac{\partial \mathcal{L}(\theta)}{\partial a_3}$.



3. Explain the different interpretations of PCA in detail.
4. Define the convolution operation for a CNN network. Give an example of a filter that we apply to an image then it captures useful information from the image.
5. If the input is of size $227 \times 227 \times 3$ and we use 40 filters of size 11×11 for the convolution layer with ~~1 zero~~ stride and zero number of padding then what is the volume of the output?
6. Why do the various weight initializations; all weight value as 0, all weight value as the same finite value, or all weight value as a small or large random value, not work in a deep neural network?

Section - C

Answer ANY TWO questions in not more than 1000 words. Each question carries 15 Points. (2 × 15 = 30 Points)

1. Explain in detail the COBOW, Word2Vec and GloVe models of predicting a word or context word.
2. Describe the working of RNN and LSTM in detail.
3. Describe all the different variants of gradient descent algorithms.